Beam Design and AI Model Comparison

Jacob Coone | Individual Project Part 2

GitHub repository: https://github.com/jakecoone/vip-beam-agent

Method

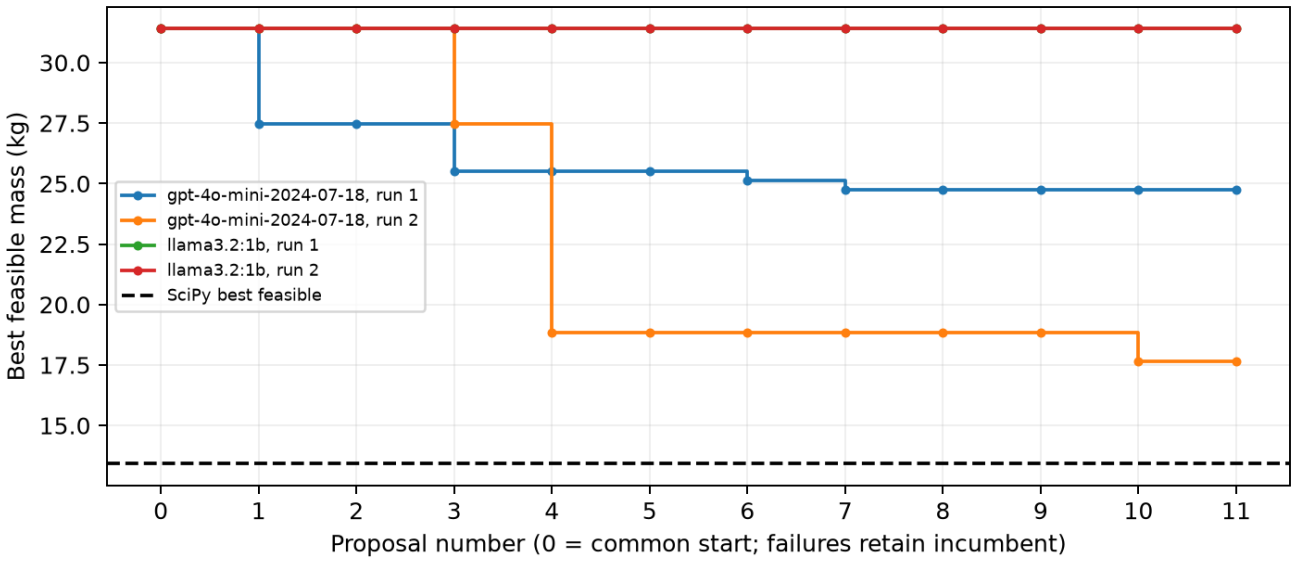
The goal was to find the lightest feasible beam using the supplied beam_model.py. Every run started at 50 x 80 mm. SciPy used SLSQP. Each AI model had two independent runs with 11 proposals each. Both models received the same prompt structure, validation rule, and their own previous results. AI designs were never chosen by a numerical optimizer.

The common start returned 31.4000 kg, 0.4883 mm deflection, and 46.8750 MPa stress. Feasible: True.

Results

Method / run	Best kg	W mm	H mm	Defl. margin mm	Stress margin MPa	Evals	Invalid	Time s
SLSQP	13.4233	20.0000	85.4988	0.0000	17.4014	38	0	0.01
GPT mini / 1	24.7275	50.0000	63.0000	0.0002	44.4142	12	0	15.42
GPT mini / 2	17.6625	30.0000	75.0000	0.0123	31.1111	12	0	12.36
Llama 1B / 1	31.4000	50.0000	80.0000	0.5117	73.1250	9	3	43.21
Llama 1B / 2	31.4000	50.0000	80.0000	0.5117	73.1250	11	1	30.46

Optimizer convergence: True. Optimization terminated successfully Returned design: 20.000000 x 85.498797 mm; strict feasibility: False; feasible within normalized 1e-6 tolerance: True. The table uses strictly feasible evaluated designs, which can differ from the final solver point.



Comparison and run variation

gpt-4o-mini-2024-07-18, run 1: 4 improvements to the best feasible mass, 1 repeated designs, 0 invalid replies (including 0 out-of-bounds replies), 0 service failures, and 0 evaluator failures. 5 evaluated proposals were infeasible. Sequential improvement is consistent with using feedback, but does not prove the model understood the beam.

gpt-4o-mini-2024-07-18, run 2: 3 improvements to the best feasible mass, 0 repeated designs, 0 invalid replies (including 0 out-of-bounds replies), 0 service failures, and 0 evaluator failures. 5 evaluated proposals were infeasible. Sequential improvement is consistent with using feedback, but does not prove the model understood the beam.

The two gpt-4o-mini-2024-07-18 runs differed by 7.0650 kg in best feasible mass. Their best mass was +4.2392 kg relative to SciPy.

llama3.2:1b, run 1: 0 improvements to the best feasible mass, 6 repeated designs, 3 invalid replies (including 1 out-of-bounds replies), 0 service failures, and 0 evaluator failures. 0 evaluated proposals were infeasible. No improvement over the starting design was observed in this run.

llama3.2:1b, run 2: 0 improvements to the best feasible mass, 9 repeated designs, 1 invalid replies (including 0 out-of-bounds replies), 0 service failures, and 0 evaluator failures. 0 evaluated proposals were infeasible. No improvement over the starting design was observed in this run.

The two llama3.2:1b runs differed by 0.0000 kg in best feasible mass. Their best mass was +17.9767 kg relative to SciPy.

Reproducibility and limitations

Temperature was 0.7; output limit was 160 tokens. No explicit random seed was set. The exact requested models were gpt-4o-mini-2024-07-18 and llama3.2:1b. Returned model identifiers are logged. Ollama used an 8192-token context and top_p=1. Strict JSON validation rejected extra keys, strings, booleans, nonfinite values, duplicate keys, markdown and out-of-bounds dimensions. Invalid replies and service errors consumed a proposal without a beam evaluation. No reply was repaired or retried.

Only two runs per model were used, so the results are a small sample. Elapsed time includes service latency and local model loading, but excludes plotting and report generation. Software, hardware, and service differences affect timing. SLSQP caches exact repeated coordinates; its evaluation count is the actual number of black-box calls. AI repeats are evaluated and counted. Margins are limit minus output; negative margins are infeasible. The worksheet contains a meters/mm inconsistency; the example and named parameters use millimeters, so this experiment uses millimeters and checks the starting outputs. This is a static, ideal beam model, not an impact or local-contact stress model.

Software: {"matplotlib": "3.11.2", "numpy": "2.5.3", "openai": "2.54.0", "platform": "macOS-26.6.2-arm64-arm-64bit-Mach-O", "python": "3.14.7", "python-dotenv": "1.2.4", "reportlab": "4.5.1", "scipy": "1.18.1"}

Evaluator SHA-256: eb88415e9837b35c84fc9689eee80788aa9830e5162c8da7fd8a8f9699583ce7

Documentation: https://developers.openai.com/api/docs/models/gpt-4o-mini ; https://docs.ollama.com/api/chat